

4(a)	straight (horizontal) lines and from the +0.90 kV plate/to the 0 V plate	B1
	(lines are) equally spaced	B1
4(b)	weight/gravitational force and electric force	B1
4(c)	$s = \frac{1}{2}at^2$ or $s = ut + \frac{1}{2}at^2$ and $u = 0$	C1
	$2.0 = \frac{1}{2} \times 9.81 \times t^2$ so $t = 0.64$ s	A1
4(d)	$0.080 = \frac{1}{2} \times a \times 0.64^2$	C1
	$a = 0.39$ m s ⁻²	A1
4(e)(i)	$E = (\Delta)V / (\Delta)d$	C1
	$E = 0.90 \times 10^3 / 0.12$ $= 7.5 \times 10^3$ N C ⁻¹	A1
4(e)(ii)	$ma = Eq$ or $F = ma$ and $F = Eq$	C1
	$q / m = 0.39 / 7.5 \times 10^3$ $= 5.2 \times 10^{-5}$ C kg ⁻¹	A1
4(f)(i)	no effect	B1
4(f)(ii)	decreases/smaller	B1